

Модуль 4. Лабораторная 3

1) Для заданной на схеме schema-lab3 сети, состоящей из управляемых коммутаторов и персональных компьютеров настроить на коммутаторах логическую топологию используя протокол IEEE 802.1Q, для передачи пакетов VLAN333 между коммутаторами использовать Native VLAN.

Настройка оставшихся PC:

```
PC4> ip 192.168.1.5/24
PC5> ip 192.168.1.6/24
PC6> ip 192.168.1.7/24
```

Создание VLAN на всех коммутаторах:

```
enable
configure terminal
vlan 20
    name VLAN20
exit
vlan 333
    name VLAN333
exit
```

Layer2Switch-1:

```
interface range GigabitEthernet0/0-3
    switchport trunk encapsulation dot1q
    switchport mode trunk
    switchport trunk native vlan 333
    switchport trunk allowed vlan 20,333
    no shutdown
exit

interface range GigabitEthernet1/0-3
    switchport trunk encapsulation dot1q
    switchport mode trunk
    switchport trunk native vlan 333
    switchport trunk allowed vlan 20,333
    no shutdown
exit

end
```

Layer2Switch-2:

```
enable
configure terminal

interface range GigabitEthernet0/0-3
    switchport trunk encapsulation dot1q
    switchport mode trunk
    switchport trunk native vlan 333
    switchport trunk allowed vlan 20,333
    no shutdown
exit

interface range GigabitEthernet1/0-3
    switchport trunk encapsulation dot1q
    switchport mode trunk
    switchport trunk native vlan 333
    switchport trunk allowed vlan 20,333
```

```
no shutdown
exit

end
```

Layer2Switch-3:

```
enable
configure terminal
interface range GigabitEthernet0/0-3
    switchport trunk encapsulation dot1q
    switchport mode trunk
    switchport trunk native vlan 333
    switchport trunk allowed vlan 20,333
    no shutdown
exit

interface GigabitEthernet1/0
    switchport mode access
    switchport access vlan 20
    no shutdown
exit

interface GigabitEthernet1/1
    switchport mode access
    switchport access vlan 333
    no shutdown
exit
```

Layer2Switch-4:

```
enable
configure terminal
interface range GigabitEthernet0/0-3
    switchport trunk encapsulation dot1q
    switchport mode trunk
    switchport trunk native vlan 333
    switchport trunk allowed vlan 20,333
    no shutdown
exit

interface GigabitEthernet1/0
    switchport mode access
    switchport access vlan 20
    no shutdown
exit

interface GigabitEthernet1/1
    switchport mode access
    switchport access vlan 333
    no shutdown
exit
```

Layer2Switch-5:

```
enable
configure terminal
interface range GigabitEthernet0/0-3
    switchport trunk encapsulation dot1q
    switchport mode trunk
    switchport trunk native vlan 333
```

```

switchport trunk allowed vlan 20,333
no shutdown
exit

interface GigabitEthernet1/0
switchport mode access
switchport access vlan 20
no shutdown
exit

interface GigabitEthernet1/1
switchport mode access
switchport access vlan 333
no shutdown
exit

```

2) Проверить доступность персональных компьютеров, находящихся в одинаковых VLAN и недоступность находящихся в различных, результаты задокументировать.

VLAN 20:

```

PC1> ping 192.168.1.4
84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=35.993 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=21.071 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=31.571 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=24.670 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=38.328 ms

PC1> ping 192.168.1.6
84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=23.327 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=44.434 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=38.045 ms
84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=28.248 ms
84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=34.528 ms

PC1> 

```

VLAN 333:

```

PC2> ping 192.168.1.5
84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=39.793 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=33.117 ms
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=30.516 ms
84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=54.998 ms
84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=48.146 ms

PC2> ping 192.168.1.7
84 bytes from 192.168.1.7 icmp_seq=1 ttl=64 time=76.300 ms
84 bytes from 192.168.1.7 icmp_seq=2 ttl=64 time=28.778 ms
84 bytes from 192.168.1.7 icmp_seq=3 ttl=64 time=19.187 ms
84 bytes from 192.168.1.7 icmp_seq=4 ttl=64 time=39.270 ms
84 bytes from 192.168.1.7 icmp_seq=5 ttl=64 time=55.894 ms

PC2> 

```

Недоступно из разных сетей:

```
PC1 - PuTTY

PC1> ping 192.168.1.3
host (192.168.1.3) not reachable

PC1> ping 192.168.1.5
host (192.168.1.5) not reachable

PC1> █
```

```
PC2 - PuTTY

PC2> ping 192.168.1.4
host (192.168.1.4) not reachable

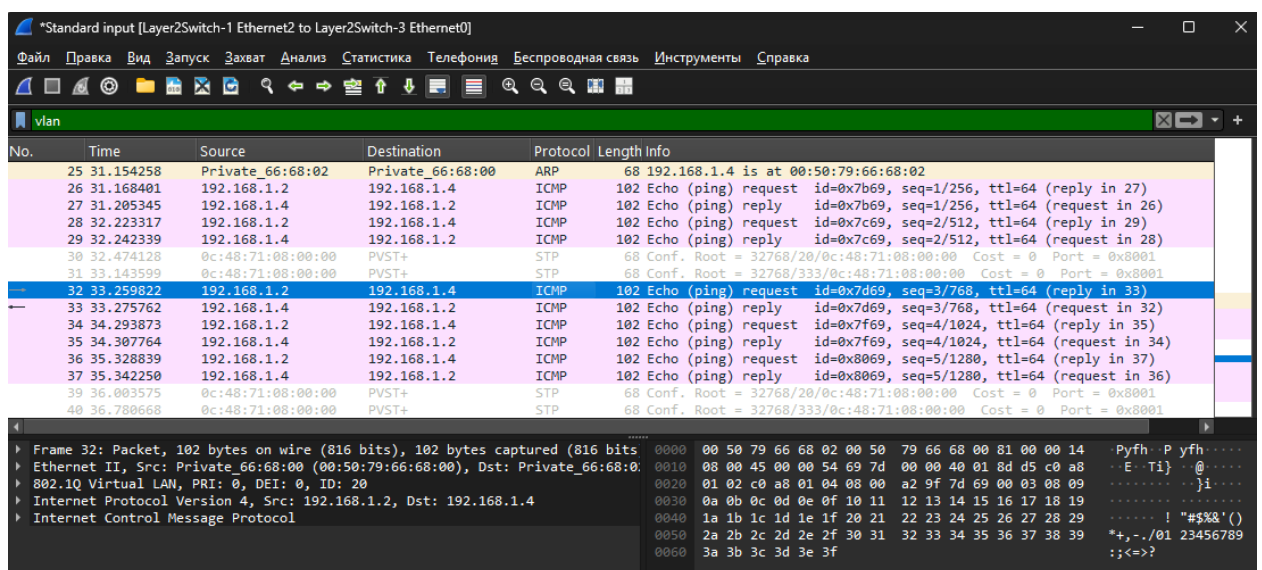
PC2> ping 192.168.1.6
host (192.168.1.6) not reachable

PC2> █
```

PC, находящиеся в пределах одного VLAN, доступны друг другу. Узлы VLAN20 успешно обмениваются ICMP-пакетами между собой, аналогично у VLAN333. При попытке обмена данными между узлами, принадлежащими различным VLAN, ICMP Echo Reply не получен.

3) Перехватить в WireShark пакеты с тегами и без тегов (nb!), результаты задокументировать.

С тегами:



Без тегов:

*Standard input [Layer2Switch-1 Ethernet2 to Layer2Switch-3 Ethernet0]						
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка						
Примените фильтр отображения ... <Ctrl-/>						
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	0c:48:71:08:00:00	PVST+	STP	68	Conf. Root = 32768/20/0c:48:71:08:00:00 Cost = 0 Port = 0x8001
2	0.828362	0c:48:71:08:00:00	PVST+	STP	68	Conf. Root = 32768/333/0c:48:71:08:00:00 Cost = 0 Port = 0x8001
3	3.446787	0c:48:71:08:00:00	PVST+	STP	68	Conf. Root = 32768/20/0c:48:71:08:00:00 Cost = 0 Port = 0x8001
4	4.196133	0c:48:71:08:00:00	PVST+	STP	68	Conf. Root = 32768/333/0c:48:71:08:00:00 Cost = 0 Port = 0x8001
5	4.323950	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.1.5? Tell 192.168.1.3
6	4.337833	Private_66:68:03	Private_66:68:01	ARP	64	192.168.1.5 is at 00:50:79:66:68:03
7	4.352383	192.168.1.3	192.168.1.5	ICMP	98	Echo (ping) request id=0x6f6c, seq=1/256, ttl=64 (reply in 8)
8	4.385497	192.168.1.5	192.168.1.3	ICMP	98	Echo (ping) reply id=0x6f6c, seq=1/256, ttl=64 (request in 7)
9	5.391499	192.168.1.3	192.168.1.5	ICMP	98	Echo (ping) request id=0x706c, seq=2/512, ttl=64 (reply in 10)
10	5.418888	192.168.1.5	192.168.1.3	ICMP	98	Echo (ping) reply id=0x706c, seq=2/512, ttl=64 (request in 9)
11	6.433420	192.168.1.3	192.168.1.5	ICMP	98	Echo (ping) request id=0x716c, seq=3/768, ttl=64 (reply in 12)
12	6.455515	192.168.1.5	192.168.1.3	ICMP	98	Echo (ping) reply id=0x716c, seq=3/768, ttl=64 (request in 11)
13	7.038957	0c:48:71:08:00:00	PVST+	STP	68	Conf. Root = 32768/20/0c:48:71:08:00:00 Cost = 0 Port = 0x8001
14	7.460209	192.168.1.3	192.168.1.5	ICMP	98	Echo (ping) request id=0x726c, seq=4/1024, ttl=64 (reply in 15)
15	7.485354	192.168.1.5	192.168.1.3	ICMP	98	Echo (ping) reply id=0x726c, seq=4/1024, ttl=64 (request in 14)
Frame 9: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:03 Internet Protocol Version 4, Src: 192.168.1.3, Dst: 192.168.1.5 Internet Control Message Protocol						
0000 00 50 79 66 68 03 00 50 79 66 68 01 08 00 45 00 Pyfh P yfh E 0010 00 54 6c 70 00 00 40 01 8a e0 c0 a8 01 03 c0 a8 Tlp @ 0020 01 05 08 00 af 9d 70 6c 00 02 08 09 0a 0b 0c 0dpl 0030 0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d ..!#\$%&'()*+,- 0040 1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d ./012345 6789:;<= 0050 2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d >? 0060 3e 3f						

При передаче трафика VLAN20 в Ethernet-кадре присутствует заголовок IEEE 802.1Q. При передаче трафика VLAN333 заголовок IEEE 802.1Q в кадре отсутствует.